



Total Marks: **60 (20 x 3)**

Time: **90 Minutes**

- Total no of questions are **four (4)**. Answer any **three (3)**.
- Figure in bracket [] next to each question indicates marks for that question.
- Write your set number on top of your answer sheet
- Use activation voltage $V_\gamma(\text{diode}) = 0.6 \text{ V}$, $V_\gamma(\text{transistor}) = 0.5 \text{ V}$, $V_{BE}(\text{forward active}) = 0.7 \text{ V}$ and $V_D(\text{conducting voltage of Diode}) = 0.7 \text{ V}$, for all the questions.
- There is no need to draw circuits in your answer sheet.

Name:	ID:	Section:
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Question 1

For the ECL circuit in **figure 1**, ignore the base currents.

[20]

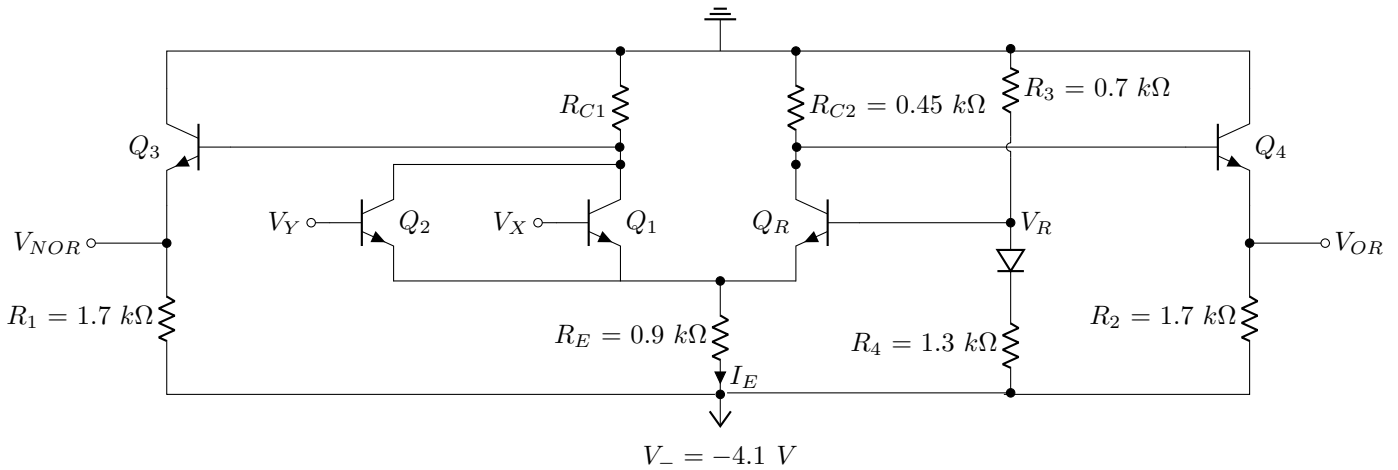


Figure 1

CO1	(a)	Calculate the reference voltage, V_R of the circuit in figure 1.	[5]
CO2	(b)	Using the reference voltage from (a) calculate the logic HIGH and logic LOW levels of V_{OR} (in volt) in figure 1. [hint: V_{OR} is low when $V_X = V_Y = \text{LOW}$. In that case current through $R_{C1} = 0$ and $I_{R_{C2}} = I_E$]	[3+7]
CO2	(c)	Calculate R_{C1} such that the logic LOW level of V_{NOR} matches that of V_{OR} .	[5]

Question 2

The triangular-wave generator circuit consists of an Integrator and a non-inverting schmitt Trigger circuit. Here, $R_i = 12\text{ K}\Omega$, $R_2 = pR = 30\text{ K}\Omega$, and $C = 0.1\text{ }\mu\text{F}$. The saturation voltages of the op-amps are $+V_{sat}(V_H) = +13\text{ V}$ and $-V_{sat}(V_L) = -13\text{ V}$.

(a)	CO2	Calculate the threshold voltages $V_{UT}(V_{TH})$ and $V_{LT}(V_{TL})$ of the Schmitt circuit.	[5]
(b)	CO5	Find the frequency of oscillation, F and the Falling and Rising time Periods, T1 and T2 of the generated triangular-wave signal. Utilizing these values, determine the duty cycle of the accompanying square wave .	[12]
(c)	CO5	Select a suitable value for the capacitor C to make the Time Period of the triangular wave 3.2ms .	[3]

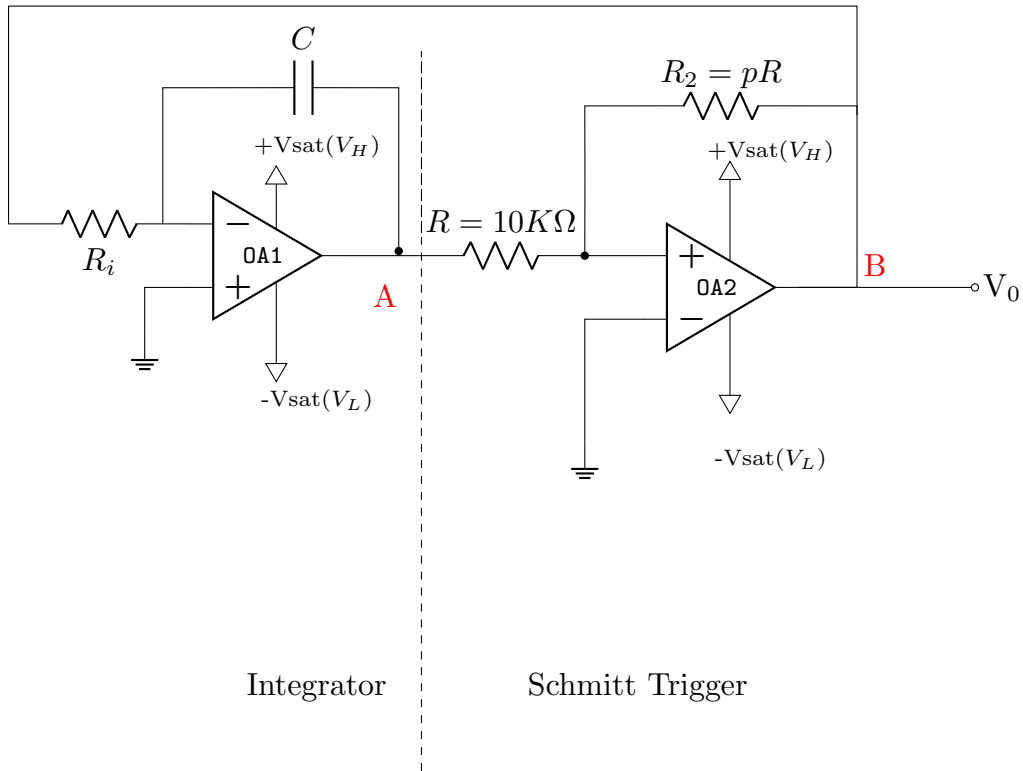


Figure 2

Question 3

Flash ADC in figure 3 has reference voltage of $V_{REF} = 8\text{ V}$. The 8-to-3 priority encoder of the ADC circuit has logical High = 4 V and logical Low = 0 V that is used to drive the inputs of the DAC of figure 4. [20]

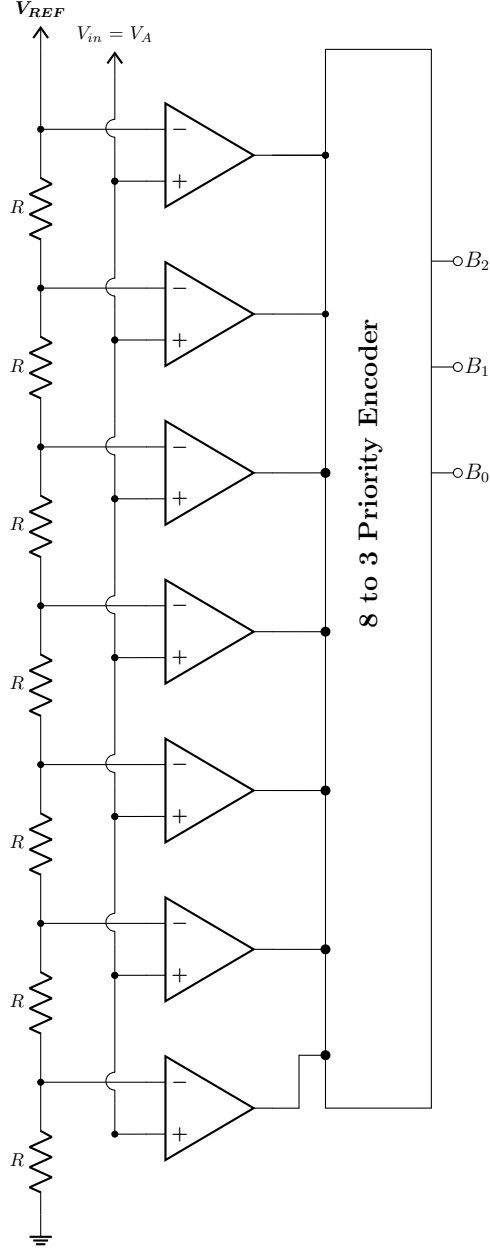


Figure 3: Flash ADC

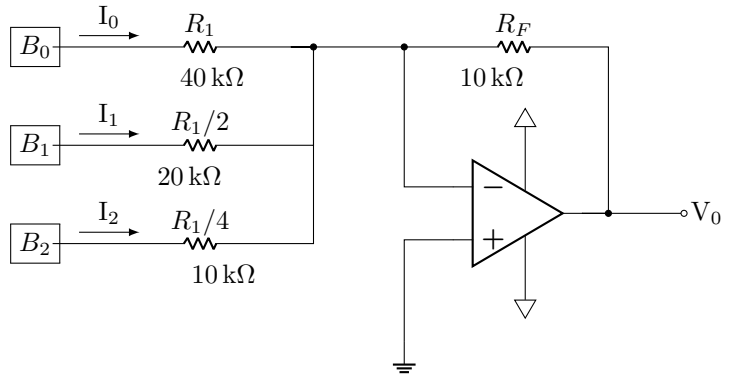


Figure 4: Binary weighted DAC

CO2	(a)	Find all the quantization voltage levels of the analog-to-digital converter circuit in figure 3.	[8]
CO3	(b)	For all 8 different input combinations, find the output voltage V_o of the digital-to-analog converter circuit in figure 4.	[8]
CO3	(c)	Suppose the ADC outputs are used to drive the inputs of the DAC. For (i) $V_{in} = 3.5\text{ V}$ and (ii) $V_{in} = 6.7\text{ V}$ calculate the binary output of the ADC and V_o of the DAC.	[4]

Question 4

The enhancement-load NMOS compound gate in **figure 5** has 2 input transistors, T_1 and T_2 with inputs , V_1 and V_2 respectively. The transistor parameters are $K_n = 0.2 \text{ mA/V}^2$, $V_{TN1} = 0.4V$, $V_{TN2} = 1V$, $V_{TNL} = 0.9V$. Also, The 2-input NMOS compound gate in **figure 5** gives an output of **logical 1** only when both inputs are **logical 0**. [20]

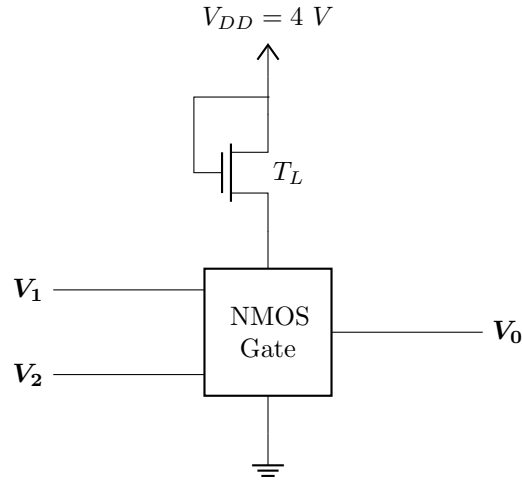


Figure 5

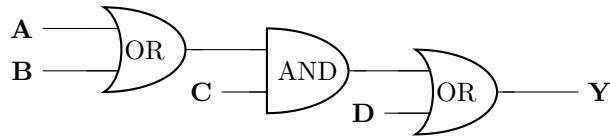


Figure 6

CO2	(a)	Find the truth table for the logic gate in figure 5 and derive the expression for output V_0	[2+2]
CO1	(b)	Find the operating mode of the load NMOS transistor in figure 5.	[4]
CO2	(c)	In figure 5 determine the value of v_0 in V for $V_1 = V_2 = 4 \text{ V}$	[4]
CO2	(d)	Design an equivalent CMOS logic circuit of figure 6. (<i>hint: derive the boolean expression for output Y and use that to design NMOS and PMOS networks</i>)	[4+4]



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- There is no need to draw circuits in your answer sheet.

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Question 1

For the rectangular wave generator in figure 1, answer the following questions.

[20]

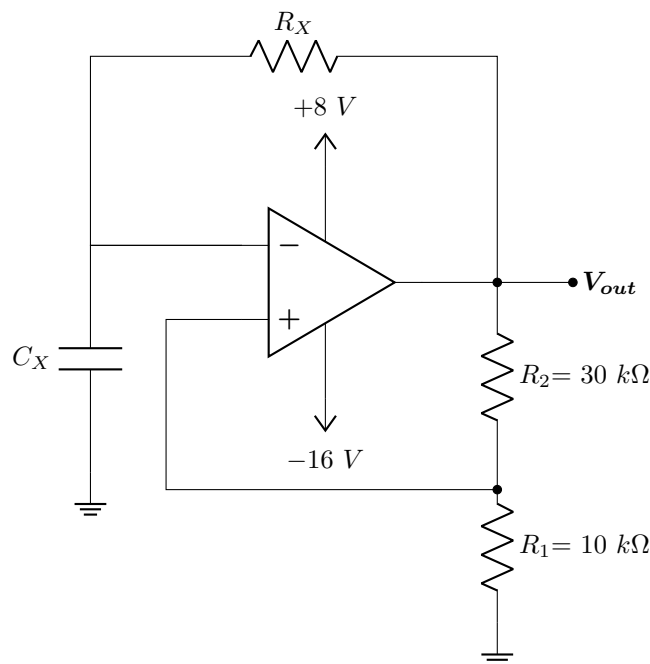


Figure 1

CO2	(a)	Identify whether the Schmitt Trigger in this circuit is inverting or non-inverting	[2]
CO2	(b)	Determine V_{TH} and V_{TL} values of the Schmitt Trigger. (<i>high and low threshold values</i>)	[4+4]
CO2	(c)	Determine the duty cycle of the rectangular wave generator circuit. (<i>hint: R_x and C_x values are not required for this part</i>)	[5]
CO2	(d)	Find the value for C_x so that the circuit has a frequency of 1000 Hz when $R_x = 910 \Omega$	[5]

Question 2

For the ECL circuit in **figure 2**, ignore the base currents.

[20]

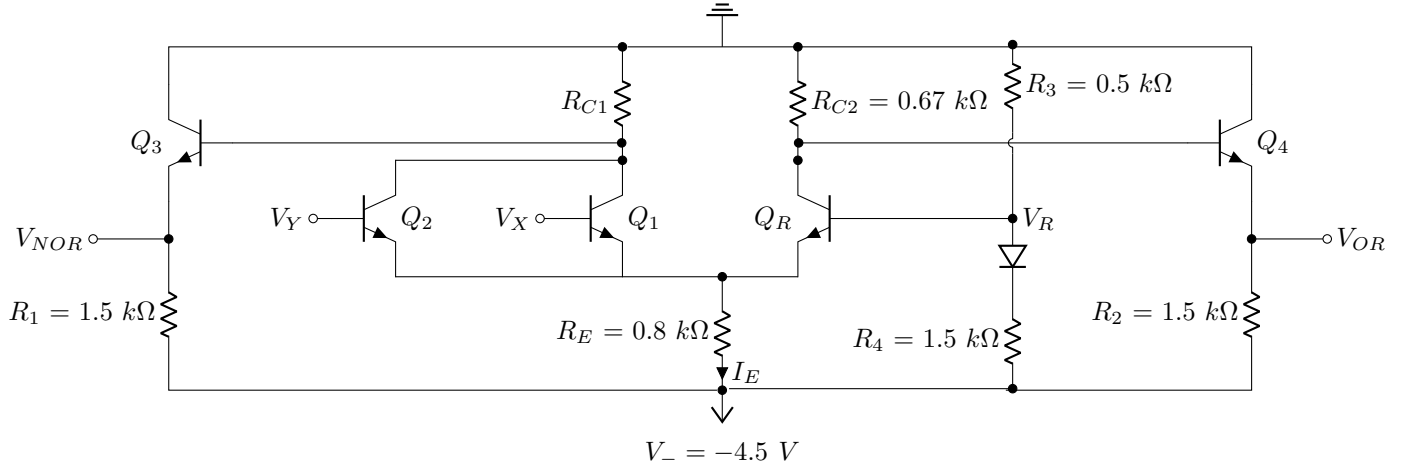


Figure 2

CO1	(a)	Calculate the reference voltage, V_R of the circuit in figure 2.	[5]
CO2	(b)	Using the reference voltage from (a) calculate the logic HIGH and logic LOW levels of V_{OR} (in volt) in figure 2. [hint: V_{OR} is low when $V_X = V_Y = LOW$. In that case current through $R_{C1} = 0$ and $I_{R_{C2}} = I_E$]	[3+7]
CO2	(c)	Calculate R_{C1} such that the logic LOW level of V_{NOR} matches that of V_{OR} .	[5]

Question 3

The enhancement-load NMOS compound gate in **figure 3** has 2 input transistors, T_1 and T_2 with inputs, V_1 and V_2 respectively. The transistor parameters are $K_n = 0.2 \text{ mA/V}^2$, $V_{TN1} = 0.4\text{V}$, $V_{TN2} = 1\text{V}$, $V_{TNL} = 0.9\text{V}$. Also, The 2-input NMOS compound gate in **figure 4** gives an output of **logical 1** only when both inputs are **logical 0**. [20]

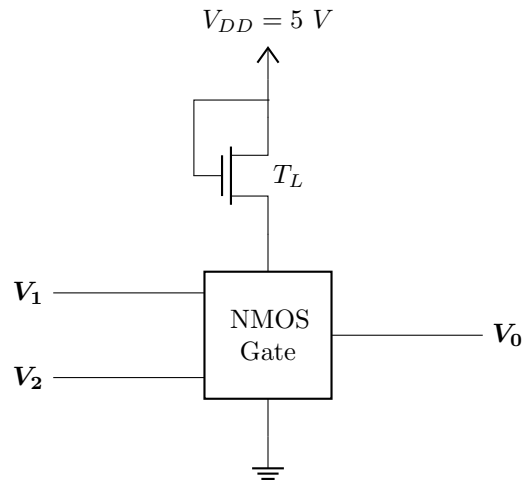


Figure 3

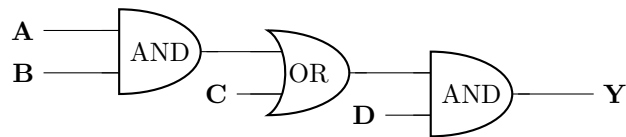


Figure 4

CO2	(a)	Find the truth table for the logic gate in figure 3 and derive the expression for output V_0	[2+2]
CO1	(b)	Find the operating mode of the load NMOS transistor in figure 3.	[4]
CO2	(c)	In figure 3 determine the value of v_0 in V for $V_1 = V_2 = 5 \text{ V}$	[4]
CO2	(d)	Design an equivalent CMOS logic circuit of figure 4. (<i>hint: derive the boolean expression for output Y and use that to design NMOS and PMOS networks</i>)	[4+4]

Question 4

Flash ADC in figure 5 has reference voltage of $V_{REF} = 16\text{ V}$. The 8-to-3 priority encoder of the ADC circuit has logical High = 8 V and logical Low = 0 V that is used to drive the inputs of the DAC of figure 6. [20]

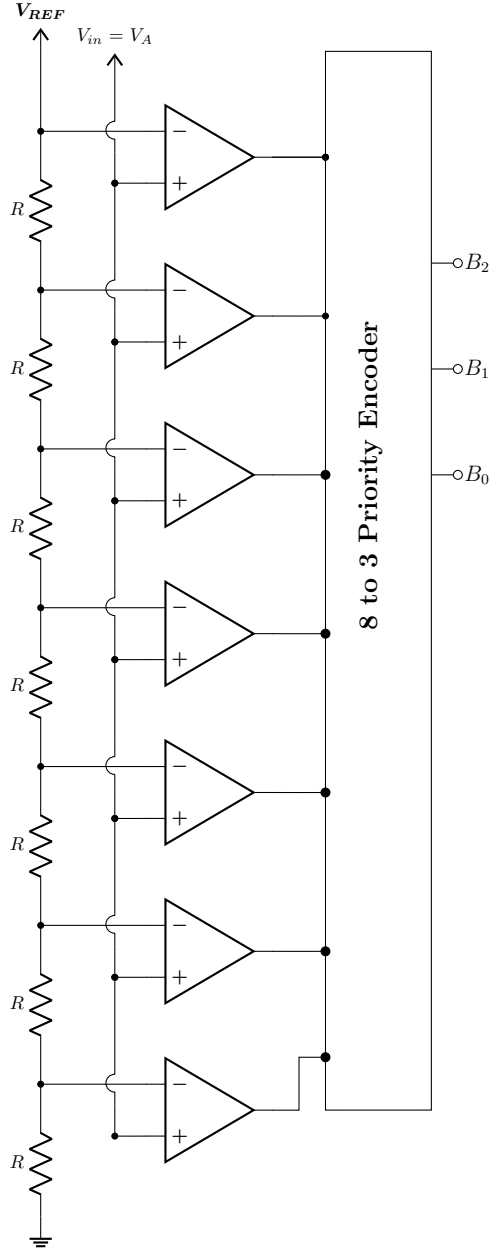


Figure 5: Flash ADC

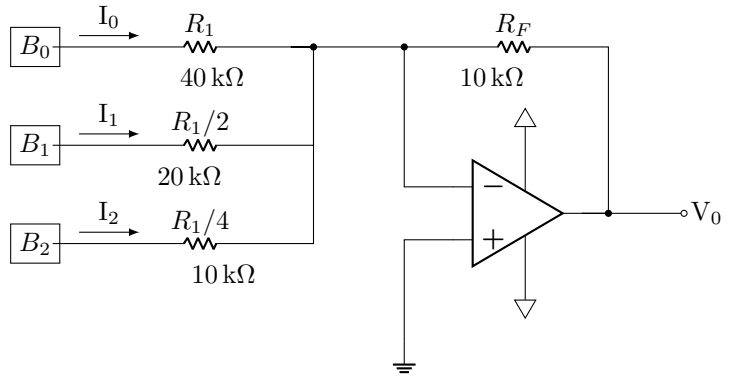


Figure 6: Binary weighted DAC

CO2	(a)	Find all the quantization voltage levels of the analog-to-digital converter circuit in figure 3.	[8]
CO3	(b)	For all 8 different input combinations, find the output voltage V_o of the digital-to-analog converter circuit in figure 4.	[8]
CO3	(c)	Suppose the ADC outputs are used to drive the inputs of the DAC. For (i) $V_{in} = 5.5\text{ V}$ and (ii) $V_{in} = 7.8\text{ V}$ calculate the binary output of the ADC and V_o of the DAC.	[4]